

Assignment 2

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1 Gradient Computation Check

To verify the correctness of the gradient computations, a small subset of the data was first selected. The forward pass and backward gradient calculations were then implemented using self-built functions. The resulting gradients were compared with those obtained from PyTorch in terms of relative errors.

The results show that the relative errors for both W and b across two layers are all below the chosen threshold $1e-6$. This indicates that the implemented gradient computations are accurate and free of bugs. The corresponding relative errors are reported in Table 1.

Table 1: Verification of gradients

Parameter	Max Relative Error
W1	4.33e-16
W2	3.60e-16
b1	3.91e-16
b2	9.06e-17

2 Training and validation Loss and Cost

This section focuses on training the network using cyclical learning rates under different hyperparameter settings. Changes of the loss, cost and accuracy on both the training and validation sets are visualized for better understanding of the model performance.

We begin with a simple setting where a single learning rate cycle is performed. Parameters are set as following: $\{\text{eta_min}=1e-5, \text{eta_max} = 1e-1, \text{n_batch} = 100, \text{lam} = 0.01, \text{n_s}=500, \text{n_cycle} = 1\}$. The test accuracy obtained through this network is 45.72%.

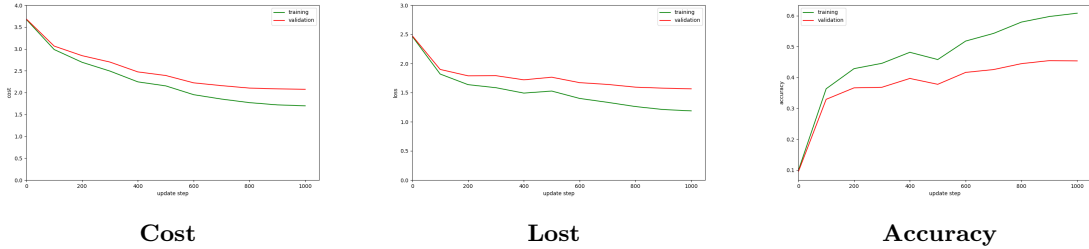


Figure 1: Curves for one-cycle of training

Then, more cycles and larger n_s are adopted. Parameters are set as following: $\{\text{eta_min}=1e-5, \text{eta_max} = 1e-1, \text{n_batch} = 100, \text{lam} = 0.01, \text{n_s}=800, \text{n_cycle} = 3\}$. The test accuracy achieved after three-cycles of training is 47.40%.

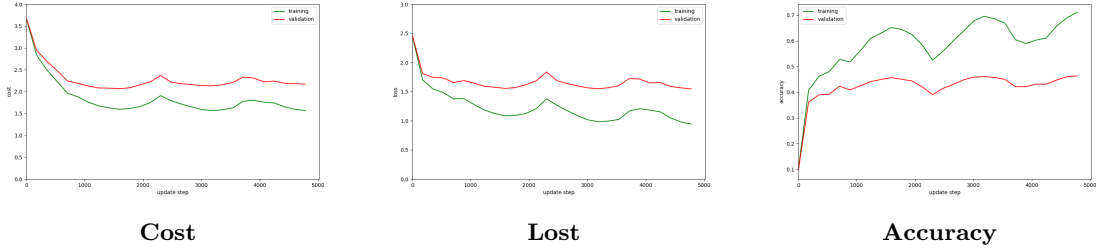


Figure 2: Curves for three-cycles of training

Comments:

- **Momentum and Fluctuation:** When training with a single learning rate cycle, both the cost and loss decrease steadily, while the accuracy shows a consistent upward trend. In contrast, with three cycles, the cost and loss curves exhibit noticeable temporary increases during their overall decline, and the accuracy curve shows distinct fluctuations with occasional drops amid its general rise.
- **Overfit:** In Fig.1, although there is a gap between the training and validation accuracies, both curves increase steadily and in sync. By contrast, in Fig.2, the training accuracy surpasses 70% by the end of training, while the validation accuracy remains around 45% after step 1600. This indicates that after the first cycle, the model’s generalization capability has largely plateaued, and further training primarily leads to overfitting.

The comparison shows that given the limited training data, too much cycles merely cause the network to overfit without improving generalization.

3 Coarse Search

The above experiments fix the regularization parameter λ at 0.01. To further improve the model performance, a coarse-to-fine random search strategy is conducted to identify better values for λ . A coarse search is first implemented. Here all five batches are used for training with 5000 samples held out as the validation set. λ are picked randomly from a log-uniform distribution over the range $[10^{-5}, 10^{-1}]$.

The remaining parameters are set as

$$\{\text{eta_min}=1\text{e-}5, \text{eta_max} = 1\text{e-}1, \text{n_s} = 900, \text{n_batch} = 100, \text{n_cycle} = 2\}$$

. Under this setting, the results of coarse search are shown in Table 2.

Table 2: Coarse Search Result

Lambda	Best Validation Accuracy
0.000010	0.5242
0.000037	0.5156
0.000139	0.5244
0.000518	0.5218
0.001931	0.5170
0.007197	0.5172
0.026827	0.4756
0.100000	0.3844

Among them, 3 best performing networks are obtained when $\lambda = 0.000139, \lambda = 0.000010, \lambda = 0.000518$.

4 Fine Search

Observing Table 2, the validation accuracy remains high around the first five values. A narrow search range is a log-uniform distribution over the range $[10^{-5.5}, 10^{-2}]$. The remaining parameters are set as

`{eta_min=1e-5, eta_max = 1e-1, n_s = 900, n_batch = 100, n_cycle =5}`

Results are shown in Table 3. Three best performing networks are obtained when $\lambda = 0.000022$, $\lambda = 0.000527$, $\lambda = 0.001000$.

Table 3: Fine Search Result

Lambda	Best Validation Accuracy
0.000003	0.5302
0.000006	0.5296
0.000011	0.5244
0.000022	0.5362
0.000041	0.5322
0.000077	0.5300
0.000147	0.5286
0.000278	0.5278
0.000527	0.5362
0.001000	0.5348

5 Test under Best Parameters

Based on the result of the fine search, the optimal value of the regularization parameter is found to be $\lambda = 0.000527$. Using this value, the network is trained on the full training dataset except for 1000 examples, which are reserved for validation. Other settings are

`{eta_min=1e-5, eta_max = 1e-1, n_s =980 , n_batch = 100, n_cycle =3}`

The training and validation loss plots are shown below.

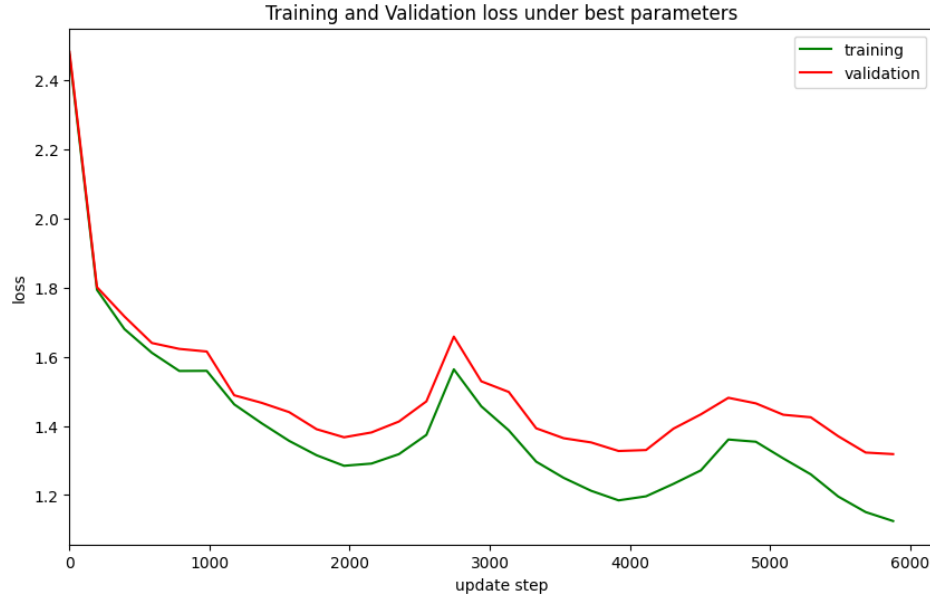


Figure 3: Loss plots under best lam

After training, the test accuracy achieved under this best lambda value is 52.01%.